

~\Downloads\Day 19.sql

```
1  -- Task 1: Employee Incremental Loading using Stream + MERGE
2  CREATE DATABASE DAY19_DB;
3
4  USE DATABASE DAY19_DB;
5
6  CREATE OR REPLACE TABLE SRC_EMPLOYEE
7  (
8      EMP_ID NUMBER,
9      EMP_NAME STRING,
10     DEPARTMENT STRING,
11     SALARY NUMBER
12 );
13
14 CREATE OR REPLACE TABLE TRG_EMPLOYEE
15 (
16     EMP_ID NUMBER,
17     EMP_NAME STRING,
18     DEPARTMENT STRING,
19     SALARY NUMBER
20 );
21
22 INSERT INTO SRC_EMPLOYEE VALUES
23 (101, 'John', 'IT', 50000),
24 (102, 'Alice', 'HR', 60000),
25 (103, 'David', 'Finance', 70000),
26 (104, 'Neha', 'Sales', 45000),
27 (105, 'Ravi', 'IT', 55000);
28
29 -- FULL LOAD
30 INSERT INTO TRG_EMPLOYEE
31 SELECT *
32 FROM SRC_EMPLOYEE;
33
34 SELECT * FROM TRG_EMPLOYEE;
35
36 SELECT * FROM SRC_EMPLOYEE;
37
38 -- CREATING STREAM
39 CREATE OR REPLACE STREAM EMP_STREAM
40 ON TABLE SRC_EMPLOYEE;
41
42 INSERT INTO SRC_EMPLOYEE VALUES
43 (106, 'Priya', 'HR', 65000),
44 (107, 'Akash', 'IT', 58000);
45
46 UPDATE SRC_EMPLOYEE
47 SET SALARY = 62000
48 WHERE EMP_ID = 101;
```

```
49
50 UPDATE SRC_EMPLOYEE
51 SET SALARY = 75000
52 WHERE EMP_ID = 103;
53
54 DELETE
55 FROM SRC_EMPLOYEE
56 WHERE EMP_ID=104;
57
58 SELECT * FROM EMP_STREAM;
59
60 -- MERGE
61 MERGE INTO TRG_EMPLOYEE T
62 USING EMP_STREAM S
63 ON T.EMP_ID = S.EMP_ID
64
65 WHEN MATCHED
66     AND S.METADATA$ACTION='DELETE'
67     AND S.METADATA$ISUPDATE=FALSE
68 THEN DELETE
69
70 WHEN MATCHED
71     AND S.METADATA$ACTION='INSERT'
72     AND S.METADATA$ISUPDATE=TRUE
73 THEN UPDATE SET
74     EMP_NAME=S.EMP_NAME,
75     DEPARTMENT=S.DEPARTMENT,
76     SALARY=S.SALARY
77
78 WHEN NOT MATCHED
79     AND S.METADATA$ACTION='INSERT'
80 THEN
81 INSERT
82 (
83 EMP_ID,
84 EMP_NAME,
85 DEPARTMENT,
86 SALARY
87 )
88 VALUES
89 (
90 S.EMP_ID,
91 S.EMP_NAME,
92 S.DEPARTMENT,
93 S.SALARY
94 );
95
96 SELECT *
97 FROM SRC_EMPLOYEE;
98
```

```
99  SELECT *
100 FROM TRG_EMPLOYEE;
101
102 SELECT *
103 FROM EMP_STREAM;
104
105
106
107
108 -- TASK 2
109 CREATE OR REPLACE TABLE SRC_PRODUCT
110 (
111  PRODUCT_ID NUMBER,
112  PRODUCT_NAME STRING,
113  CATEGORY STRING,
114  PRICE NUMBER
115 );
116
117 CREATE OR REPLACE TABLE TRG_PRODUCT
118 (
119  PRODUCT_ID NUMBER,
120  PRODUCT_NAME STRING,
121  CATEGORY STRING,
122  PRICE NUMBER
123 );
124
125 INSERT INTO SRC_PRODUCT VALUES
126 (1, 'Laptop', 'Electronics', 65000),
127 (2, 'Keyboard', 'Electronics', 1500),
128 (3, 'Mouse', 'Electronics', 800),
129 (4, 'Chair', 'Furniture', 4500),
130 (5, 'Table', 'Furniture', 9000),
131 (6, 'Bottle', 'Kitchen', 500);
132
133 INSERT INTO TRG_PRODUCT
134 SELECT *
135 FROM SRC_PRODUCT;
136
137 CREATE OR REPLACE STREAM PRODUCT_STREAM
138 ON TABLE SRC_PRODUCT;
139
140 ALTER WAREHOUSE COMPUTE_WH RESUME;
141
142 CREATE OR REPLACE TASK PRODUCT_LOAD_TASK
143 WAREHOUSE = COMPUTE_WH
144 SCHEDULE= '1 MINUTE'
145 AS
146
147 INSERT INTO TRG_PRODUCT
148
```

```
149 SELECT
150 PRODUCT_ID,
151 PRODUCT_NAME,
152 CATEGORY,
153 PRICE
154
155 FROM PRODUCT_STREAM
156
157 WHERE METADATA$ACTION='INSERT'
158 AND METADATA$ISUPDATE=FALSE;
159
160 ALTER TASK PRODUCT_LOAD_TASK
161 RESUME;
162
163 INSERT INTO SRC_PRODUCT VALUES
164 (7,'Monitor','Electronics',12000),
165 (8,'Printer','Electronics',9000),
166 (9,'Headphone','Electronics',2500);
167
168 SELECT *
169 FROM TRG_PRODUCT;
170
171 SELECT *
172 FROM TABLE
173 (
174 INFORMATION_SCHEMA.TASK_HISTORY
175 (
176 TASK_NAME=>'PRODUCT_LOAD_TASK'
177 )
178 );
179
180 SELECT *
181 FROM PRODUCT_STREAM;
```